

G1 Probability — Lesson Plan

IGCSE Mathematics 0580 | Concept Guide

G1_Q1: Basic Probability

Hook

Imagine flipping a coin. Heads or tails? You know there's a 50-50 chance — but what does that '50%' really mean? Probability is the language of uncertainty, used everywhere from weather forecasts to sports predictions to board games. Once you master the basics, you'll be able to calculate exactly how likely (or unlikely) any event is.

Prerequisites

- Understand fractions, decimals, and percentages and how to convert between them
- Be comfortable simplifying fractions to their lowest terms

Core Explanation

What is Probability?

Probability measures how likely something is to happen. It is always a number between 0 and 1 (or 0% to 100%).

- A probability of 0 means the event is impossible.
- A probability of 1 means the event is certain.
- A probability of 0.5 (or $\frac{1}{2}$ or 50%) means the event is equally likely to happen or not happen.

The Basic Formula

The probability of an event is calculated as:

$$P(\text{event}) = (\text{Number of favourable outcomes}) / (\text{Total number of possible outcomes})$$

This formula only works when every outcome is equally likely — like rolling a fair dice or drawing a card from a well-shuffled deck.

Key Terms

- **Outcome**: A single possible result (e.g., rolling a 4 on a dice)
- **Event**: A set of one or more outcomes you are interested in (e.g., rolling an even number)
- **Sample space**: The complete list of all possible outcomes
- **Equally likely**: Outcomes that have the same chance of occurring

Probability Scale

The probability scale runs from 0 to 1. Common benchmark probabilities:

- 0: Impossible (e.g., rolling a 7 on a standard dice)
- 0.25: Unlikely (e.g., drawing a heart from a deck of cards)
- 0.5: Even chance (e.g., getting heads on a coin flip)
- 0.75: Likely (e.g., not drawing a heart from a deck of cards)
- 1: Certain (e.g., the sun will rise tomorrow)

Complementary Events

If an event A has probability $P(A)$, then the probability that A does NOT happen is:

$$P(\text{not } A) = 1 - P(A)$$

This is called the complement rule. For example, if the chance of rain is 0.3, the chance of no rain is 0.7.

Expressing Probability

Probability can be written as:

- A fraction: $\frac{1}{6}$
- A decimal: 0.1667
- A percentage: 16.67%

All three forms are acceptable in IGCSE answers unless the question specifies a format.

Worked Example

****Problem:**** A bag contains 3 red marbles, 5 blue marbles, and 2 green marbles. One marble is chosen at random. What is the probability that it is blue?

****Step 1:**** Find the total number of marbles (total possible outcomes).

$$\text{Total} = 3 + 5 + 2 = 10 \text{ marbles}$$

****Step 2:**** Find the number of favourable outcomes (blue marbles).

$$\text{Number of blue marbles} = 5$$

****Step 3:**** Apply the probability formula.

$$P(\text{blue}) = \frac{5}{10} = \frac{1}{2}$$

****Answer:**** The probability of choosing a blue marble is $\frac{1}{2}$ (or 0.5, or 50%).

Common Mistakes

X Forgetting to simplify the fraction. Writing $\frac{5}{10}$ instead of $\frac{1}{2}$.

OK Fix: Always simplify your probability fractions to their lowest terms. $\frac{5}{10}$ simplifies to $\frac{1}{2}$ by dividing numerator and denominator by 5.

X Writing probability as a ratio like '1:2' instead of a fraction, decimal, or percentage.

OK Fix: Probability must be expressed as a number between 0 and 1 (fraction, decimal, or percentage). A ratio

X Counting outcomes incorrectly by not including all possibilities in the total.

OK Fix: Always double-check that you've counted every possible outcome. Write out the sample space if you're

Summary

- Probability is a number between 0 (impossible) and 1 (certain) that measures how likely an event is.
- The basic formula is $P(\text{event}) = (\text{favourable outcomes}) / (\text{total outcomes})$, assuming equally likely outcomes.
- The complement rule: $P(\text{not } A) = 1 - P(A)$.
- Probability can be expressed as a fraction, decimal, or percentage — simplify fractions where possible.
- Always check that you've correctly identified both the favourable and total number of outcomes.

G1_Q2: Sample Space Diagrams

Hook

Roll two dice and add the numbers. What's the most likely total? Is 7 really more common than 12? The answer lies in the sample space — a complete map of every possible outcome. Sample space diagrams (like two-way tables) turn messy probability problems into a simple visual grid, making it easy to spot patterns and calculate probabilities.

Prerequisites

- Understand basic probability: $P(\text{event}) = \text{favourable} / \text{total outcomes}$
- Be able to list outcomes systematically without missing any

Core Explanation

What is a Sample Space?

A sample space is the set of all possible outcomes of an experiment. For simple experiments you can list outcomes directly. For experiments with two or more stages (like rolling two dice), a sample space diagram helps you organise all possibilities.

Two-Way Tables (Sample Space Diagrams)

When you have two independent events (e.g., rolling two dice, spinning two spinners, flipping two coins), you can use a two-way table. One event's outcomes go across the top row, the other's go down the first column. Each cell in the table represents one combined outcome.

Example: Rolling Two Dice

Here is the sample space when rolling two fair six-sided dice:

	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

There are $6 \times 6 = 36$ equally likely outcomes.

Finding Probabilities from a Sample Space

Once you have the diagram, count the cells that match your event and divide by the total number of cells.

- $P(\text{sum} = 7)$: Count cells where the two numbers add to 7. These are (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 outcomes. So $P(\text{sum} = 7) = 6/36 = 1/6$.
- $P(\text{double})$: Count cells where both numbers are the same: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6) — 6 outcomes. So $P(\text{double}) = 6/36 = 1/6$.

Systematic Listing

Even without a table, you can list outcomes systematically. For example, flipping two coins:

{ HH, HT, TH, TT }

This has 4 equally likely outcomes.

The Multiplication Principle

If an experiment has two stages and the first stage has m outcomes and the second has n outcomes, the total number of combined outcomes is $m \times n$.

- Flipping a coin (2 outcomes) and rolling a dice (6 outcomes): $2 \times 6 = 12$ outcomes
- Choosing a shirt (4 options) and trousers (3 options): $4 \times 3 = 12$ outfits

This extends to three or more stages: multiply the number of outcomes at each stage.

Worked Example

****Problem:**** A coin is flipped and a fair six-sided dice is rolled. Draw a sample space diagram and find the probability of getting a head and an even number.

****Step 1:**** Identify all possible outcomes for each item.

Coin: H, T (2 outcomes)

Dice: 1, 2, 3, 4, 5, 6 (6 outcomes)

****Step 2:**** Create the sample space diagram.

	H	T
1	H,1	T,1
2	H,2	T,2
3	H,3	T,3
4	H,4	T,4
5	H,5	T,5
6	H,6	T,6

Total outcomes = $2 \times 6 = 12$

****Step 3:**** Identify favourable outcomes.

We need H and an even number: outcomes are (H,2), (H,4), (H,6). That's 3 outcomes.

****Step 4:**** Calculate the probability.

$P(\text{head and even}) = \frac{3}{12} = \frac{1}{4}$

****Answer:**** The probability is $\frac{1}{4}$ (or 0.25, or 25%).

Common Mistakes

X Forgetting that (H,T) and (T,H) are different outcomes when flipping two coins.

OK Fix: When order matters, treat each distinct pair as a separate outcome. (H,T) means coin 1 is heads, coin 2 is tails.

X Counting the total outcomes incorrectly — e.g., thinking rolling two dice has 36 but adding instead of multiplying.

OK Fix: Use the multiplication principle: if stage 1 has m outcomes and stage 2 has n outcomes, the total combined outcomes are $m \times n$.

X Missing some outcomes when listing manually, especially with three or more items.

OK Fix: Always use a systematic approach: a two-way table for two items, or a tree diagram for three or more items.

Summary

- A sample space is the complete set of all possible outcomes of an experiment.
- Two-way (sample space) tables are excellent for organising outcomes of two-stage experiments.
- The total number of combined outcomes = $m \times n$ where m and n are outcomes at each stage.
- To find a probability, count favourable outcomes in the sample space and divide by the total.
- Always list or diagram outcomes systematically to avoid missing any.

G1_Q3: Tree Diagrams

Hook

What if you flip a coin three times in a row? What's the chance of getting exactly two heads? As experiments get more stages, listing every outcome gets messy. Tree diagrams solve this problem beautifully — each branch shows a possible path, and by multiplying along branches you can find the probability of even the most complex multi-stage events.

Prerequisites

- Understand basic probability and the multiplication principle
- Be able to work with fractions and multiply them
- Understand independent events (outcomes of one stage don't affect another)

Core Explanation

What is a Tree Diagram?

A tree diagram shows all possible outcomes of a multi-stage experiment. Each 'branch' represents one possible result at each stage. By following branches from start to finish, you can trace every possible sequence of outcomes.

Structure of a Tree Diagram

- Start at the left with a point (the 'root')
- From the root, draw one branch for each possible outcome of the first stage
- Label each branch with the outcome and its probability
- From the end of each branch, draw branches for the second stage
- Continue for as many stages as needed
- The end of each final branch shows a complete outcome path

The Multiplication Rule

To find the probability of a specific sequence of events (a 'path'), multiply the probabilities along that path.

$$P(\text{path}) = P(\text{stage 1 outcome}) \times P(\text{stage 2 outcome}) \times P(\text{stage 3 outcome}) \times \dots$$

Example: Flipping a Coin Twice

First flip: two branches — H (0.5) and T (0.5)

From each, second flip: two branches — H (0.5) and T (0.5)

The four paths and their probabilities:

- HH: $0.5 \times 0.5 = 0.25$

- HT: $0.5 \times 0.5 = 0.25$
- TH: $0.5 \times 0.5 = 0.25$
- TT: $0.5 \times 0.5 = 0.25$

The sum of all path probabilities equals 1: $0.25 + 0.25 + 0.25 + 0.25 = 1$

The Addition Rule for Events with Multiple Paths

If you want the probability of an event that can happen through multiple different paths, ADD the probabilities of those paths.

$$P(\text{getting exactly one head in two flips}) = P(\text{HT}) + P(\text{TH}) = 0.25 + 0.25 = 0.5$$

Probabilities on Branches

- For independent events, each stage uses the same probabilities regardless of previous outcomes
- For experiments without replacement (e.g., drawing cards from a deck without putting them back), the probabilities change after each stage
- The probabilities on branches coming from the same point must always add up to 1

Worked Example

****Problem:**** A bag contains 3 red sweets and 2 yellow sweets. A sweet is taken at random, eaten, and then another sweet is taken. Draw a tree diagram and find the probability of getting one red and one yellow sweet in any order.

****Step 1:**** Label the first stage.

First sweet: $P(\text{red}) = 3/5$, $P(\text{yellow}) = 2/5$

****Step 2:**** Label the second stage. Since the sweet is not replaced, the probabilities change.

If first was red (3 red, 2 yellow): $P(\text{red}) = 2/4$, $P(\text{yellow}) = 2/4$

If first was yellow (3 red, 1 yellow): $P(\text{red}) = 3/4$, $P(\text{yellow}) = 1/4$

****Step 3:**** Draw and fill the tree diagram.

First branch: Red ($3/5$)

- Second branch: Red ($2/4$), Yellow ($2/4$)

First branch: Yellow ($2/5$)

- Second branch: Red ($3/4$), Yellow ($1/4$)

****Step 4:**** Calculate path probabilities.

Red then Yellow: $(3/5) \times (2/4) = 6/20 = 3/10$

Yellow then Red: $(2/5) \times (3/4) = 6/20 = 3/10$

****Step 5:**** Add the probabilities of the favourable paths.

$$P(\text{one red, one yellow}) = 3/10 + 3/10 = 6/10 = 3/5$$

****Answer:**** The probability is $\frac{3}{5}$ (or 0.6, or 60%).

Common Mistakes

X Forgetting to adjust probabilities for 'without replacement' scenarios — using the same probability

OK Fix: When items are not replaced, always recalculate the probabilities for the next stage based on what remains.

X Adding path probabilities when you should multiply, or multiplying when you should add.

OK Fix: Remember: multiply ALONG a path (for AND), add ACROSS paths (for OR). $P(A \text{ then } B) = \text{multiply}$. $P(A \text{ or } B) = \text{add}$.

X Not checking that all branch probabilities from a single point sum to 1.

OK Fix: Always verify that the probabilities on branches coming from the same node add up to 1. If they don't, there's a mistake.

Summary

- Tree diagrams show all possible outcomes of multi-stage experiments, with each path representing one sequence of events.
- Multiply probabilities along a branch to find the probability of that specific sequence.
- Add probabilities of different paths when an event can occur in multiple ways.
- For 'without replacement' problems, update probabilities at each stage based on what remains.
- The sum of all final path probabilities should always equal 1.

G1_Q4: Conditional Probability

Hook

If you know that a dice roll resulted in an even number, what's the chance it was a 6? You've just narrowed down the possibilities — that's conditional probability in action. It's about updating your probability estimate when you have extra information. This idea powers everything from medical testing to spam filters.

Prerequisites

- Understand basic probability concepts and notation
- Understand tree diagrams and how to read probabilities from them
- Be comfortable working with fractions

Core Explanation

What is Conditional Probability?

Conditional probability answers the question: 'Given that something has happened, what is the probability that something else happens?'

Notation

$P(A | B)$ means 'the probability of event A happening, given that event B has already happened'.

For example, $P(\text{rain} | \text{cloudy})$ means 'the probability of rain given that it is cloudy'. This is likely higher than the ordinary chance of rain on any random day.

The Conditional Probability Formula

$$P(A | B) = P(A \text{ and } B) / P(B)$$

Provided that $P(B)$ is not zero. In words: the probability of A given B equals the probability of both A and B happening, divided by the probability of B.

Intuition

When we know B has happened, B becomes our new 'sample space'. We only consider outcomes where B occurs. Out of those, we want the ones where A also occurs.

$$P(A | B) = (\text{outcomes where both A and B occur}) / (\text{outcomes where B occurs})$$

Conditional Probability from Tree Diagrams

Tree diagrams naturally show conditional probabilities. When you move from one stage to the next, the probabilities on the second-stage branches are conditional on what happened in the first stage.

For example, if you draw two cards without replacement:

- $P(\text{second card is an Ace} | \text{first card was an Ace}) = 3/51$ (only 3 aces left out of 51 cards)

Independence

Two events are independent if knowing one happened gives no information about the other.

If A and B are independent: $P(A | B) = P(A)$

This also means: $P(A \text{ and } B) = P(A) \times P(B)$

If A and B are dependent: $P(A | B)$ is NOT equal to $P(A)$, and $P(A \text{ and } B) = P(A) \times P(B | A)$ or $P(B) \times P(A | B)$

Worked Example

****Problem:**** In a class of 30 students, 18 study Biology and 12 study Chemistry. 8 students study both Biology and Chemistry. A student is chosen at random. Given that the student studies Chemistry, what is the probability they also study Biology?

****Step 1:**** Identify what you know.

$$P(\text{Chemistry}) = 12/30$$

$$P(\text{Biology and Chemistry}) = 8/30$$

Step 2: Apply the conditional probability formula.

$$\begin{aligned}P(\text{Biology} \mid \text{Chemistry}) &= P(\text{Biology and Chemistry}) / P(\text{Chemistry}) \\&= (8/30) / (12/30) \\&= 8/12 \\&= 2/3\end{aligned}$$

Step 3: Interpret the result.

Out of the 12 students who study Chemistry, 8 also study Biology. So 8 out of 12 = 2/3.

Answer: $P(\text{Biology} \mid \text{Chemistry}) = 2/3$.

Common Mistakes

X Confusing $P(A \mid B)$ with $P(B \mid A)$. These are generally different values.

OK Fix: Read the notation carefully. $P(A \mid B)$ asks 'given B happened, what's the chance of A?' while $P(B \mid A)$ is the other way round.

X Forgetting that $P(B)$ is the new denominator and using the original total number of outcomes instead.

OK Fix: When calculating $P(A \mid B)$, restrict your attention to only the outcomes where B occurs. The denominator is the number of outcomes where B occurs, not the total number of outcomes.

X Assuming independence without checking — treating dependent events as independent.

OK Fix: Always check whether events are independent. If knowing one event gives you information about the other, they are dependent.

Summary

- Conditional probability $P(A \mid B)$ is the probability of A occurring given that B has already occurred.
- The formula is $P(A \mid B) = P(A \text{ and } B) / P(B)$, where $P(B)$ is not zero.
- Given B, the sample space shrinks to only outcomes where B occurs.
- Two events are independent if $P(A \mid B) = P(A)$, or equivalently $P(A \text{ and } B) = P(A) \times P(B)$.
- Tree diagrams with 'without replacement' scenarios naturally show conditional probabilities on the later branches.

G1_Q5: Venn Diagrams and Probability

Hook

Venn diagrams are the visual superheroes of probability. When you have overlapping groups — students who play football, students who play tennis, and some who do both — a Venn diagram shows instantly how the numbers relate. No confusion, no double-counting, just a clear picture of who belongs where and the probabilities that follow.

Prerequisites

- Understand basic probability: $P(\text{event}) = \text{favourable} / \text{total}$
- Understand the notation for sets: intersection (AND), union (OR), complement (NOT)
- Be able to interpret simple Venn diagrams

Core Explanation

Venn Diagrams and Probability

A Venn diagram uses overlapping circles to show relationships between sets. Each region represents a different combination of membership.

Key Notation

- $P(A)$: Probability of event A
- $P(A \cap B)$: Probability of A AND B (the intersection — where sets overlap)
- $P(A \cup B)$: Probability of A OR B (the union — everything in either set)
- $P(A')$: Probability of NOT A (the complement)
- $P(A \text{ only})$: Probability of A but NOT B

Structure of a Two-Set Venn Diagram

Draw two overlapping circles inside a rectangle. The rectangle represents the entire sample space. The circles represent events A and B.

- Region 1: A only (inside A, outside B)
- Region 2: B only (inside B, outside A)
- Region 3: Both A and B (the overlapping area, $A \cap B$)
- Region 4: Neither A nor B (outside both circles, inside the rectangle)

The total of all four regions equals the total number of items (or total probability = 1).

Probability Formulas from Venn Diagrams

$$- P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This is the addition rule. We subtract $P(A \cap B)$ because it's counted twice (once in $P(A)$ and once in $P(B)$).

$$- P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$- P(A \text{ only}) = P(A) - P(A \cap B)$$

$$- P(A') = 1 - P(A)$$

Mutually Exclusive Events

If two events CANNOT happen at the same time (e.g., rolling a 3 and rolling a 5 on a single dice), they are mutually exclusive. Their circles do NOT overlap.

For mutually exclusive events: $P(A \cap B) = 0$, so $P(A \cup B) = P(A) + P(B)$

Using Venn Diagrams for Conditional Probability

You can also find conditional probabilities from Venn diagrams:

$$P(A | B) = P(A \cap B) / P(B) = (\text{number in overlap}) / (\text{number in B circle total})$$

Worked Example

****Problem:**** In a survey of 50 people, 30 like tea, 20 like coffee, and 12 like both tea and coffee. Draw a Venn diagram and find:

- P(like tea only)
- P(like tea or coffee)
- P(like neither)

****Step 1:**** Draw a Venn diagram with two circles: Tea and Coffee.

****Step 2:**** Fill in the numbers.

Both (Tea $\cap$ Coffee): 12

Tea only: $30 - 12 = 18$

Coffee only: $20 - 12 = 8$

Neither: $50 - (18 + 12 + 8) = 50 - 38 = 12$

****Step 3:**** Calculate the probabilities.

a) $P(\text{Tea only}) = 18/50 = 9/25$

b) $P(\text{Tea or Coffee}) = P(\text{Tea}) + P(\text{Coffee}) - P(\text{both}) = 30/50 + 20/50 - 12/50 = 38/50 = 19/25$

Or directly from the diagram: $(18 + 12 + 8)/50 = 38/50 = 19/25$

c) $P(\text{neither}) = 12/50 = 6/25$

****Answer:**** a) $9/25$, b) $19/25$, c) $6/25$.

Common Mistakes

X Putting the 'both' number directly into the overlap without first subtracting it from the individual counts.

OK Fix: The value in the overlap is the number who belong to BOTH. For 'A only', subtract the overlap from A's total.

X Adding $P(A)$ and $P(B)$ without subtracting the overlap when finding $P(A \cup B)$.

OK Fix: Always use $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If you don't subtract the overlap, you count the intersection twice.

X Forgetting to include the 'neither' region in the Venn diagram or in the total count.

OK Fix: The rectangle represents the entire sample space. Make sure you account for items that belong to neither set.

Summary

- Venn diagrams visually show relationships between sets using overlapping circles inside a rectangle.
- The intersection $P(A \cap B)$ is the overlap; the union $P(A \cup B)$ includes everything in either circle.
- The addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

- Mutually exclusive events have no overlap: $P(A \cap B) = 0$.
- Always fill Venn diagrams starting from the overlap (both), then work outward to 'only' regions, then 'neither'.

G1_Q6: Expected Value

Hook

If you roll a dice 600 times, how many sixes would you 'expect' to get? About 100. You can't predict any single roll, but over many trials, the pattern becomes clear. Expected value is the bridge between probability and real-world prediction — it tells you the average outcome over many repetitions, and it's the key concept behind insurance, gambling, and business forecasting.

Prerequisites

- Understand basic probability: $P(\text{event}) = \text{favourable} / \text{total}$
- Be comfortable multiplying fractions or decimals by whole numbers
- Understand the concept of 'average' or 'mean'

Core Explanation

What is Expected Value?

Expected value is the average result you would expect if you repeated an experiment many, many times. It doesn't tell you what will happen on any single trial — it tells you the long-run average.

The Basic Formula

For a single event (e.g., number of successes in n trials):

Expected number = Probability of success $\times$ Number of trials

Or: $E = P(\text{event}) \times n$

Example: Rolling a Dice 120 Times

$P(\text{getting a 6}) = 1/6$

Expected number of sixes in 120 rolls = $(1/6) \times 120 = 20$

This doesn't mean you'll get exactly 20 sixes — you might get 18 or 23. But over many sets of 120 rolls, the average would be close to 20.

Expected Value with Different Outcomes and Values

Sometimes each outcome has a different 'value' (e.g., points in a game, money won/lost). The expected value is:

$$E = (\text{value1} \times P1) + (\text{value2} \times P2) + \dots + (\text{valuen} \times Pn)$$

Where each value is multiplied by its probability, and you sum them all up.

Example: A Prize Game

A game costs 1 to play. You spin a spinner:

- 1/4 chance of winning 2
- 1/4 chance of winning 1
- 1/2 chance of winning nothing

What is the expected winnings?

$$E = (2 \times 1/4) + (1 \times 1/4) + (0 \times 1/2)$$

$$E = 0.50 + 0.25 + 0 = 0.75$$

The expected winnings are 0.75 per game. Since it costs 1 to play, the expected profit is -0.25. On average, you lose 25 cents per game.

Expected Value and Fairness

A game is 'fair' if the expected value equals the cost to play. In the example above, the game is unfair because the expected return (0.75) is less than the cost (1).

To make it fair, the expected value should equal the cost.

Applications

Expected value is used in:

- Insurance: Calculating premiums based on expected claims
- Gambling: Casinos design games so expected value is slightly in their favour
- Business: Estimating expected profit or loss from different decisions
- Quality control: Expected number of defective items in a batch

Worked Example

****Problem:**** A bag contains 4 red marbles and 6 blue marbles. You draw a marble at random, record its colour, and put it back. If you do this 50 times, how many red marbles do you expect to draw?

****Step 1:**** Find the probability of drawing a red marble.

$$P(\text{red}) = 4 / (4 + 6) = 4/10 = 2/5 = 0.4$$

****Step 2:**** Apply the expected value formula.

$$\text{Expected number} = P(\text{red}) \times \text{number of trials}$$

$$E = 0.4 \times 50 = 20$$

****Step 3:**** Interpret the result.

Over 50 draws (with replacement), you'd expect to get approximately 20 red marbles.

****Answer:**** The expected number of red marbles is 20.

Common Mistakes

X Thinking expected value means you WILL get exactly that number every time.

OK Fix: Expected value is the long-run average, not a guaranteed outcome. Actual results will vary around the

X Forgetting to multiply the probability by the number of trials — just stating the probability as th

OK Fix: Expected number = probability x number of trials. If the probability is 0.2 and you have 50 trials, the ex

X Confusing expected value with the most likely outcome, especially in skewed distributions.

OK Fix: The expected value is a weighted average. The most likely single outcome might be different. For exam

Summary

- Expected value is the long-run average result when an experiment is repeated many times.
- For counting successes: $E = P(\text{event}) \times \text{number of trials}$.
- For outcomes with different values: $E = \text{sum of (value} \times \text{probability) for each outcome}$.
- A game is 'fair' when the expected return equals the cost to play.
- Expected value predicts the average over many trials, not the result of any single trial.

G1_Q7: Experimental vs Theoretical Probability

Hook

In theory, a coin should land heads half the time. But flip it 10 times, and you might get 7 heads. Flip it 1000 times, and the proportion gets closer to 50%. This is the tug-of-war between theoretical probability (what should happen) and experimental probability (what actually happens). Understanding the difference — and why they converge — is at the heart of statistics.

Prerequisites

- Understand basic probability concepts and calculations
- Understand expected value and relative frequency
- Be comfortable working with fractions and proportions

Core Explanation

Theoretical Probability

Theoretical probability is what you calculate using logic and mathematics, assuming all outcomes are equally likely.

$P(\text{theoretical}) = (\text{number of favourable outcomes}) / (\text{total number of possible outcomes})$

It does NOT require any experiments or data. It's the 'ideal' probability based on the structure of the situation.

Example: $P(\text{heads on a fair coin}) = 1/2$. This is a theoretical probability — we know it without flipping any coins.

Experimental Probability (Relative Frequency)

Experimental probability is based on actual data from trials or experiments. It's calculated as:

$P(\text{experimental}) = (\text{number of times the event occurred}) / (\text{total number of trials})$

This is also called relative frequency.

Example: You flip a coin 50 times and get 28 heads. The experimental probability of heads is $28/50 = 0.56$.

The Law of Large Numbers

The Law of Large Numbers says that as the number of trials increases, the experimental probability gets closer and closer to the theoretical probability.

- With 10 coin flips: experimental might be 0.7 or 0.3 — quite far from 0.5
- With 100 coin flips: experimental will likely be between 0.45 and 0.55
- With 1000 coin flips: experimental will be very close to 0.5

This doesn't mean outcomes 'even out' in the short term. It means the average over many trials becomes more stable and predictable.

When to Use Each

- **Theoretical probability**: Use when you know the structure of the situation and all outcomes are equally likely (dice, cards, coins, fair spinners)
- **Experimental probability**: Use when you don't know the underlying probabilities, or when you want to check if something is fair (e.g., is this dice biased? Does a medical treatment work?)

Estimating Probability from Experiments

If you can't calculate a theoretical probability (e.g., the chance it will rain tomorrow, the chance a new product will be successful), you can conduct trials and use the experimental probability as an estimate.

For IGCSE, you may be asked to:

- Calculate experimental probabilities from data
- Compare experimental and theoretical probabilities

- Predict how many trials are needed for experimental probability to get close to theoretical
- Use experimental probability to estimate the number of favourable outcomes in a larger sample

Worked Example

****Problem:**** A spinner with 4 equal sections (numbered 1-4) is spun 200 times. The number 3 comes up 62 times.

- What is the theoretical probability of landing on 3?
- What is the experimental probability of landing on 3?
- If the spinner is fair, roughly how many 3s would you expect in 200 spins? Does the data suggest the spinner is fair?

****Step 1:**** Find the theoretical probability.

$$P(3) = 1/4 = 0.25$$

****Step 2:**** Find the experimental probability.

$$P(3) = 62/200 = 31/100 = 0.31$$

****Step 3:**** Calculate the expected number.

$$\text{Expected} = 0.25 \times 200 = 50$$

The actual result was 62, which is 12 more than expected. This is higher than typical random variation — it might suggest the spinner is biased towards 3, but more trials would be needed to be sure.

****Answer:**** a) $1/4$, b) $31/100$ (0.31), c) Expected 50; the experimental result of 62 is notably higher, suggesting possible bias.

Common Mistakes

X Assuming that a small number of trials should give exactly the theoretical probability.

OK Fix: With a small number of trials, experimental results can vary widely from theoretical expectations. The Law of Large Numbers states that experimental probability approaches theoretical probability as the number of trials increases.

X Claiming a dice or spinner is biased based on too few trials.

OK Fix: A few unusual results don't prove bias. You need many trials (hundreds or thousands) to have confidence in your results.

X Using the wrong denominator when calculating experimental probability — using the number of favourable outcomes.

OK Fix: Experimental probability = (number of times event occurred) / (total number of trials). The denominator should be the total number of trials, not the number of favourable outcomes.

Summary

- Theoretical probability is calculated from known equally-likely outcomes: favourable / total.
- Experimental probability (relative frequency) comes from actual data: occurrences / trials.
- The Law of Large Numbers states that experimental probability approaches theoretical probability as the number of trials increases.
- Use theoretical probability for fair, structured situations. Use experimental probability when the structure is unknown or you want to test fairness.

- A single small experiment cannot prove a probability — always consider the number of trials when interpreting experimental results.